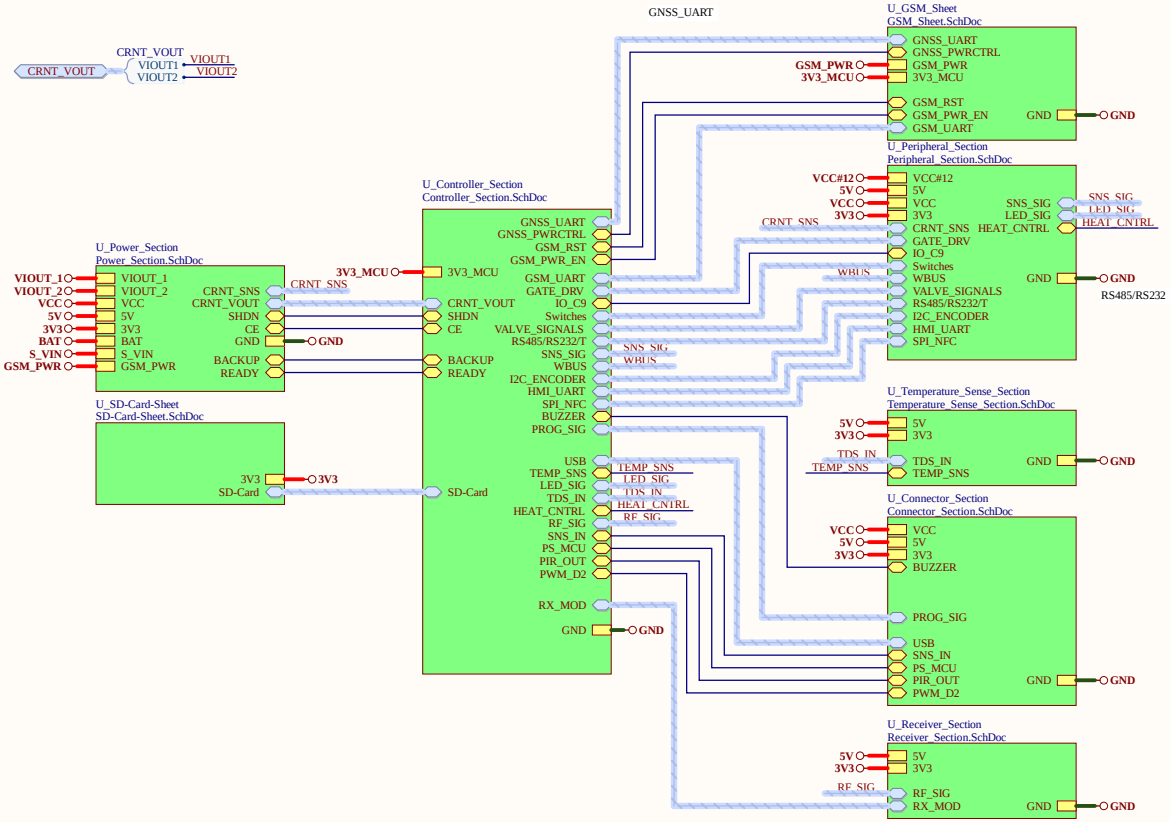
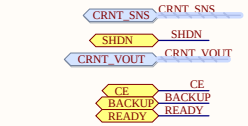
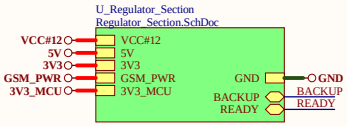
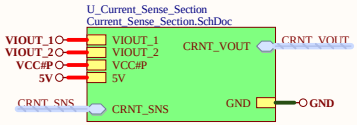


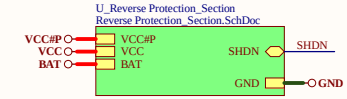
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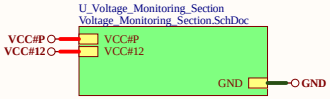
Title: Top_Sheet.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 17

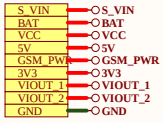


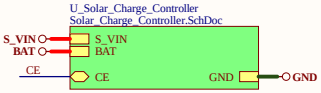








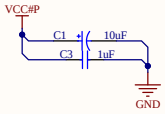




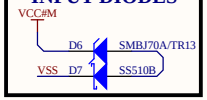
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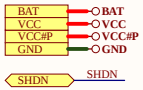
Title: Power_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 17

DECOUPLING CAPACITOR

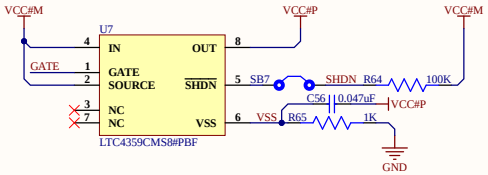


INPUT DIODES

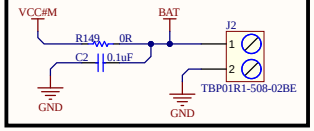




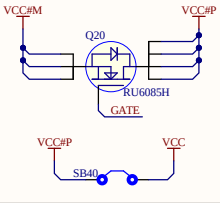
REVERSE POLARITY PROTECTION IC



BATTERY / MAIN INPUT



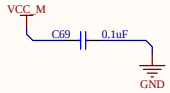
MOSFET



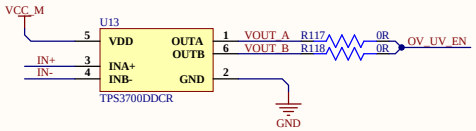
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Title: Reverse Protection_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 17

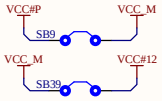
DECOUPLING CAPACITOR



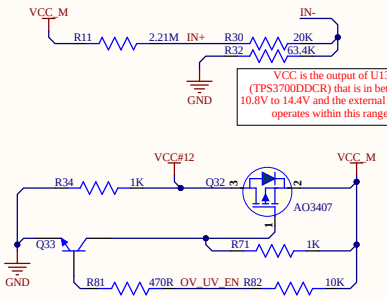
UNDER AND OVER VOLTAGE PROTECTION IC

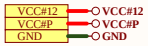


VOLTAGE TO MONITOR

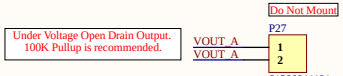


SUPPORTING CIRCUITRY

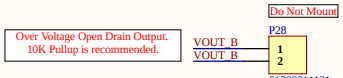




UNDER VOLTAGE OUTPUT

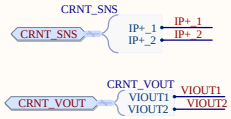


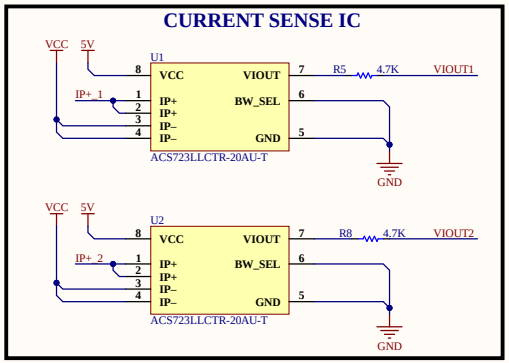
OVER VOLTAGE OUTPUT



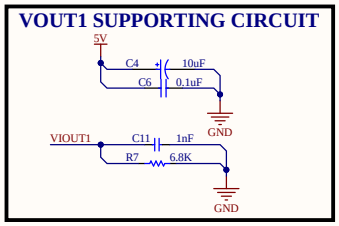
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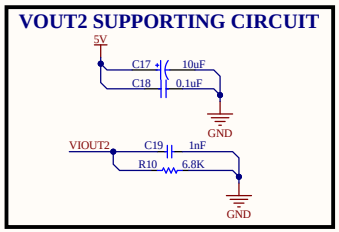
Title: Voltage_Monitoring_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 17

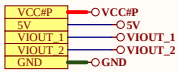




Two current sensing IC's are used to provide support to monitor current of two Pumps. Whereas currently we have only one pump in our V3 Controller.



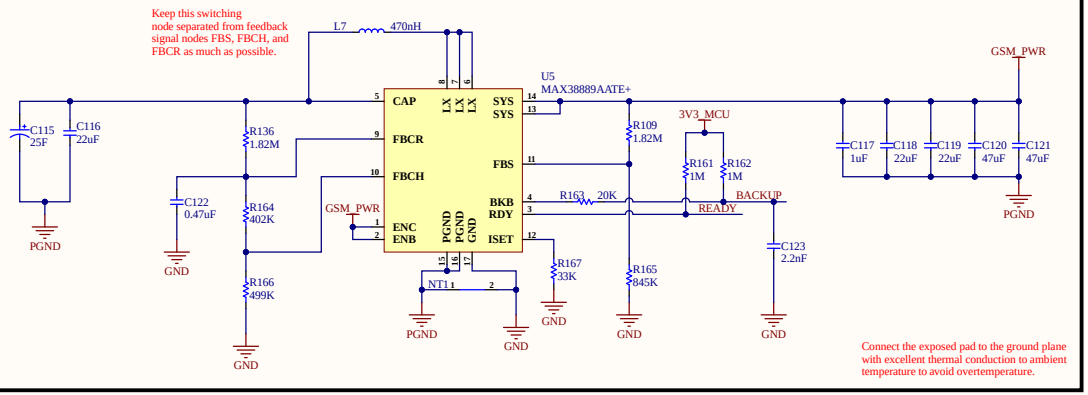




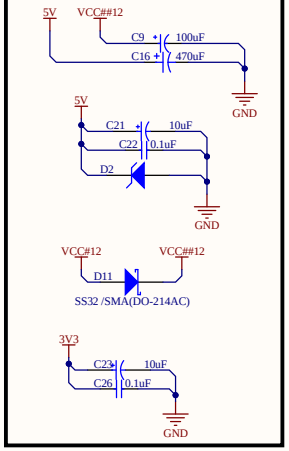
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Title: Current_Sense_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 17

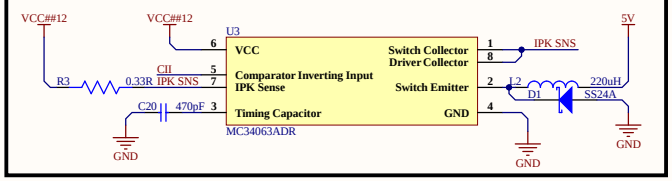
SUPERCAPACITOR BACKUP CONTROLLER



DECOUPLING CAPACITOR

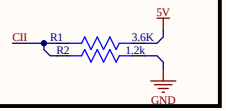


5V BUCK REGULATOR

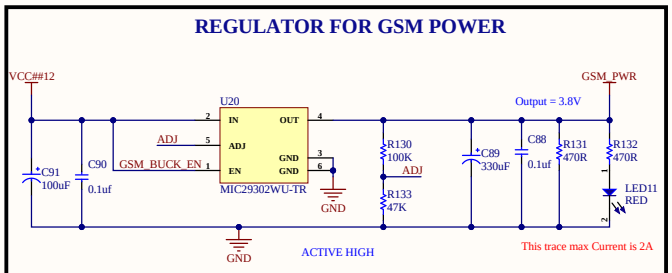


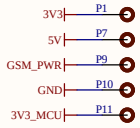
VOUT = 1.25 (1+ R2/R1)
R2=3.6K
R1=1.2K

SUPPORTING CIRCUIT

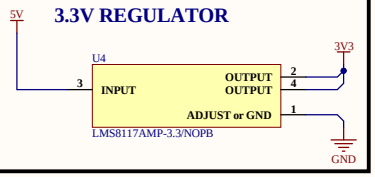


REGULATOR FOR GSM POWER

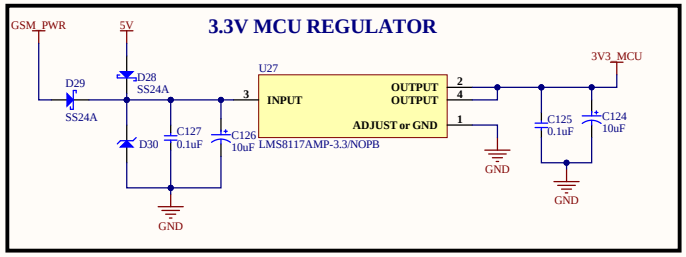


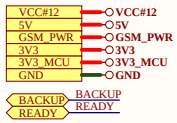


3.3V REGULATOR



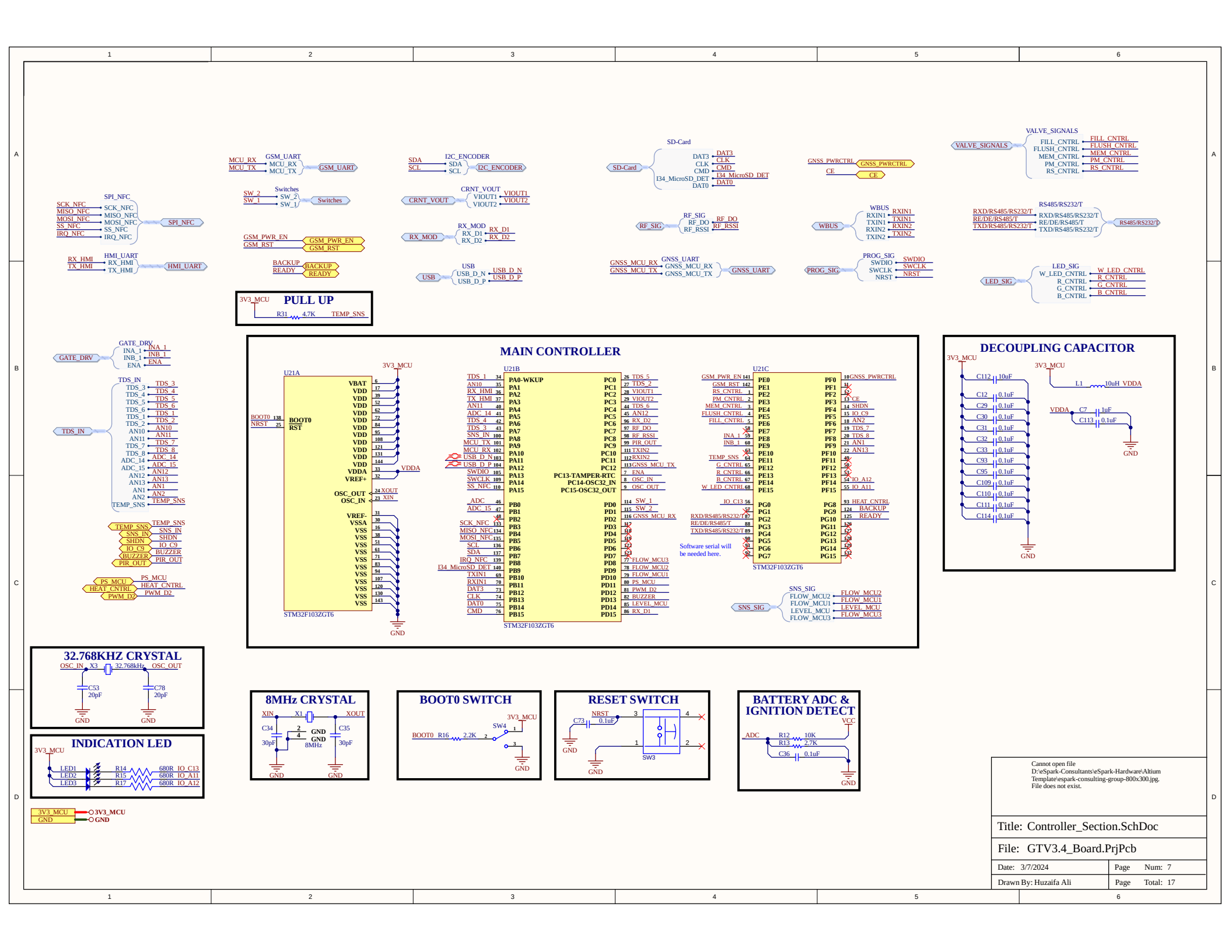
3.3V MCU REGULATOR





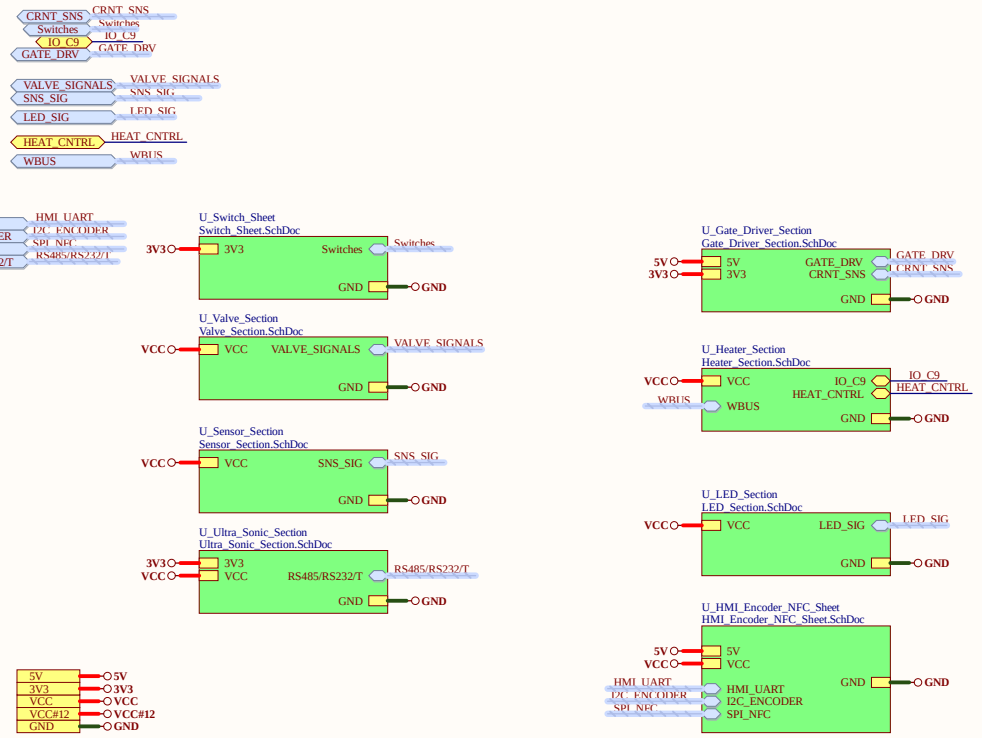
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Title: Regulator_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 17



Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Controller_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 17

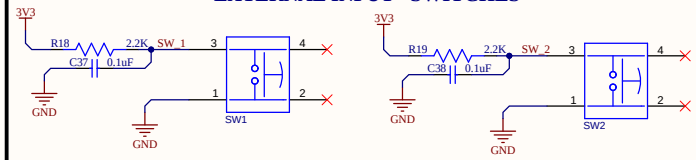


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File does not exist.

Title: Peripheral_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 17

SW_1
SW_2
Switches
SW_1
SW_2
Switches

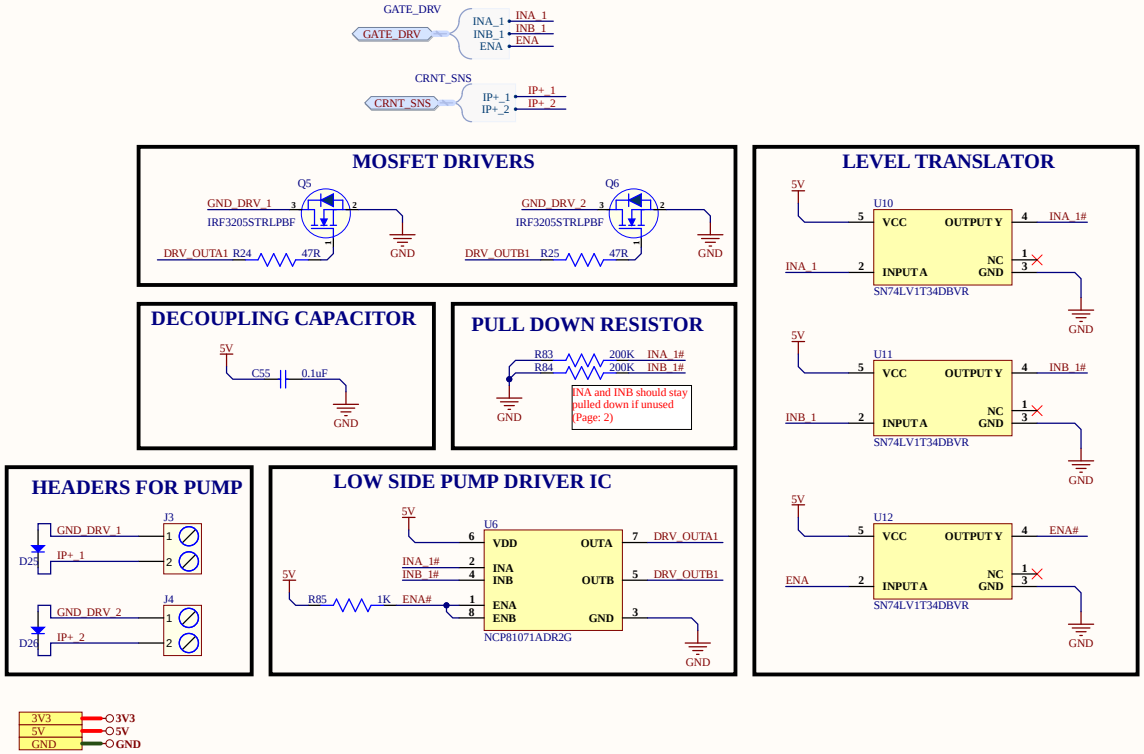
EXTERNAL INPUT - SWITCHES



3V3
GND
3V3
GND

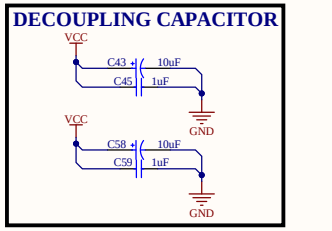
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Title: Switch_Sheet.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 9
Drawn By: Huzaifa Ali	Page Total: 17



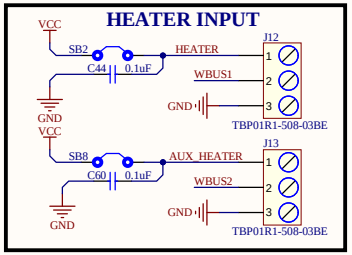
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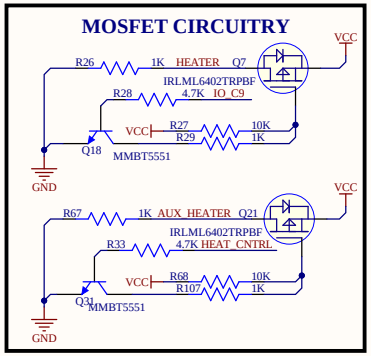
Title: Gate_Driver_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 10
Drawn By: Huzaifa Ali	Page Total: 17

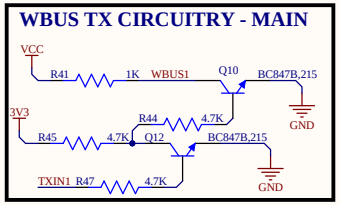


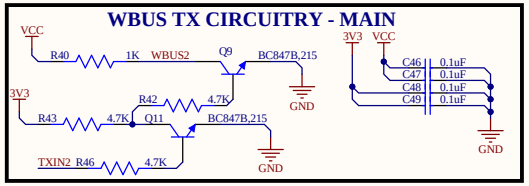


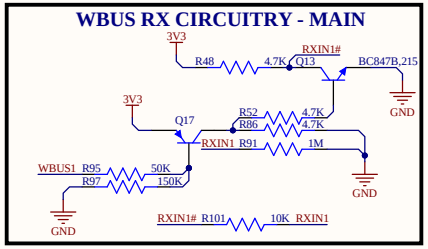


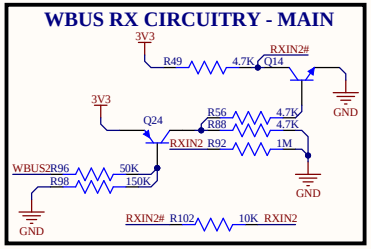


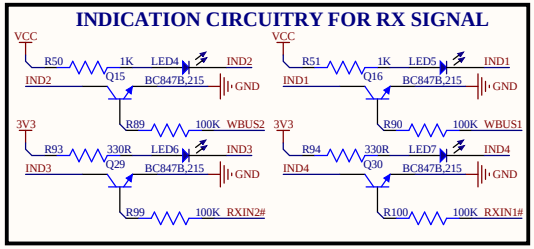






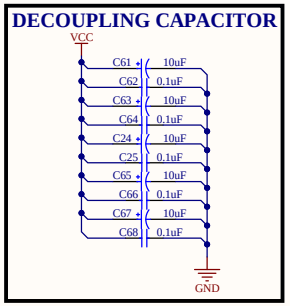


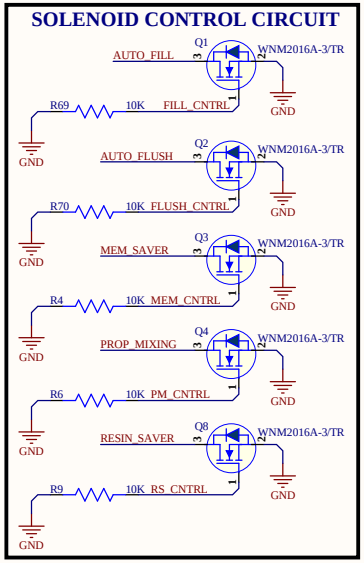


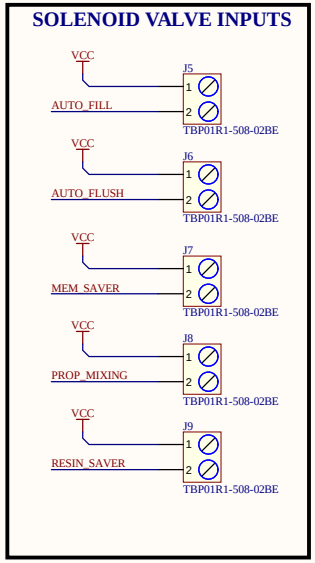


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Title: Heater_Section.SchDoc		
File: GTV3.4_Board.PrjPcb		
Date: 3/7/2024	Page	Num: 11
Drawn By: Huzaifa Ali	Page	Total: 17







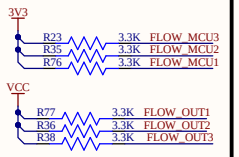




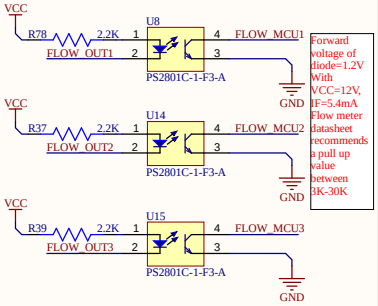
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Title: Valve_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 12
Drawn By: Huzaifa Ali	Page Total: 17

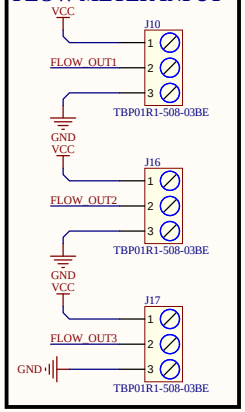
PULL UP RESISTORS



OPTO ISOLATION FOR FLOW METER



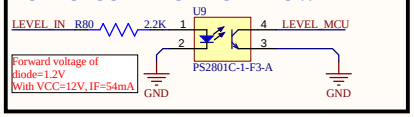
FLOW METER INPUT



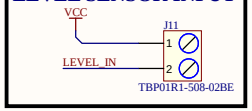
SUPPORTING CIRCUIT

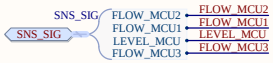


OPTO ISOLATION FOR FLOW METER



LEVEL SENSOR INPUT

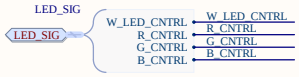


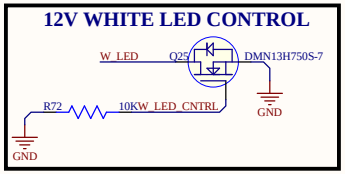


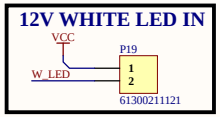


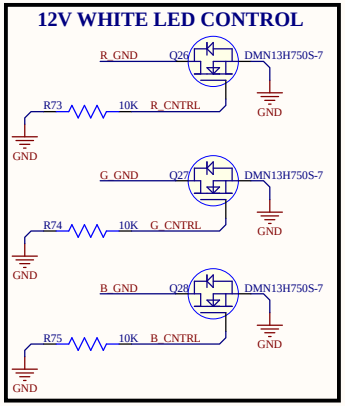
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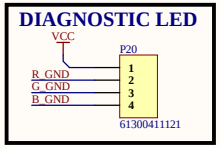
Title: Sensor_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 13
Drawn By: Huzaifa Ali	Page Total: 17









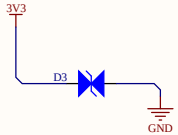




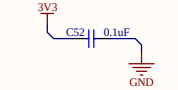
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Title: LED_Section.SchDoc	
File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 14
Drawn By: Huzaifa Ali	Page Total: 17

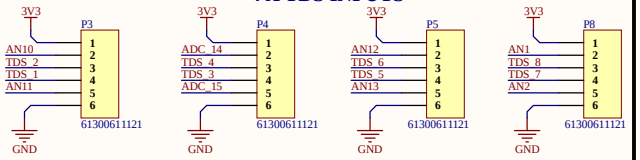
PULL UP RESISTORS



DECOUPLING CAP

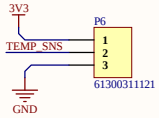


4 X TDS INPUTS

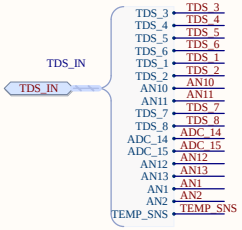


These TDS headers connect with TDS header on V3 Controller PCB via Female to Female Cable.

TEMP SENSOR DS18B20

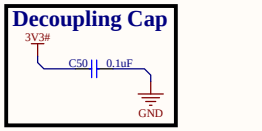


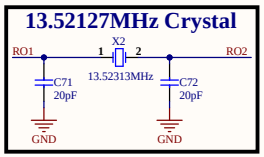


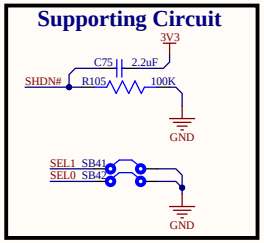


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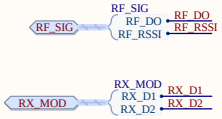
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File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 15
Drawn By: Huzaifa Ali	Page Total: 17

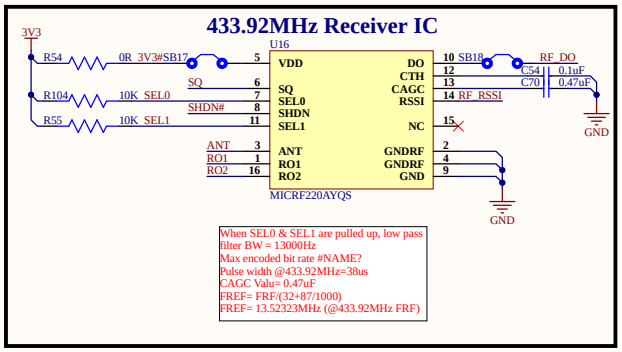


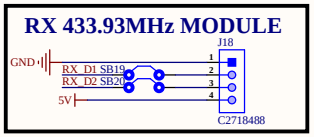


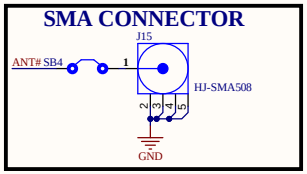


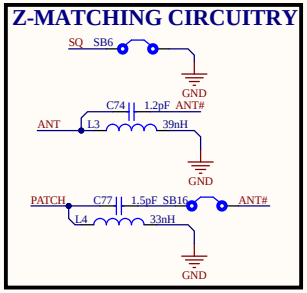






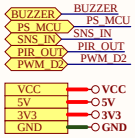


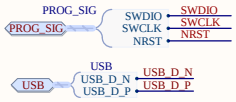


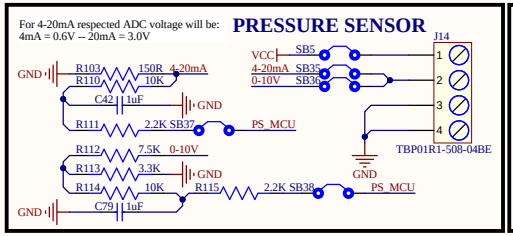


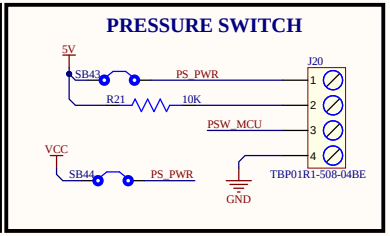
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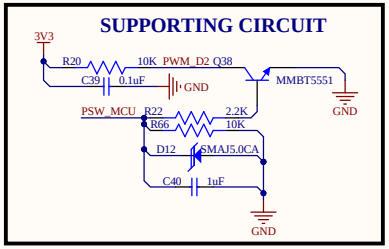
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Date: 3/7/2024	Page Num: 16
Drawn By: Huzaifa Ali	Page Total: 17

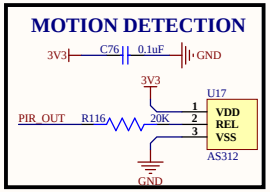


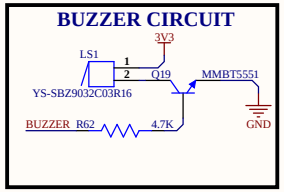


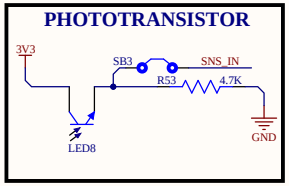


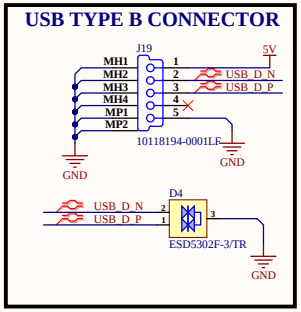


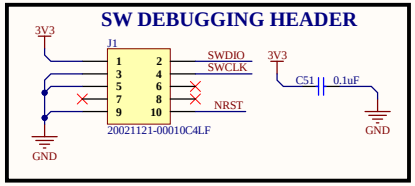








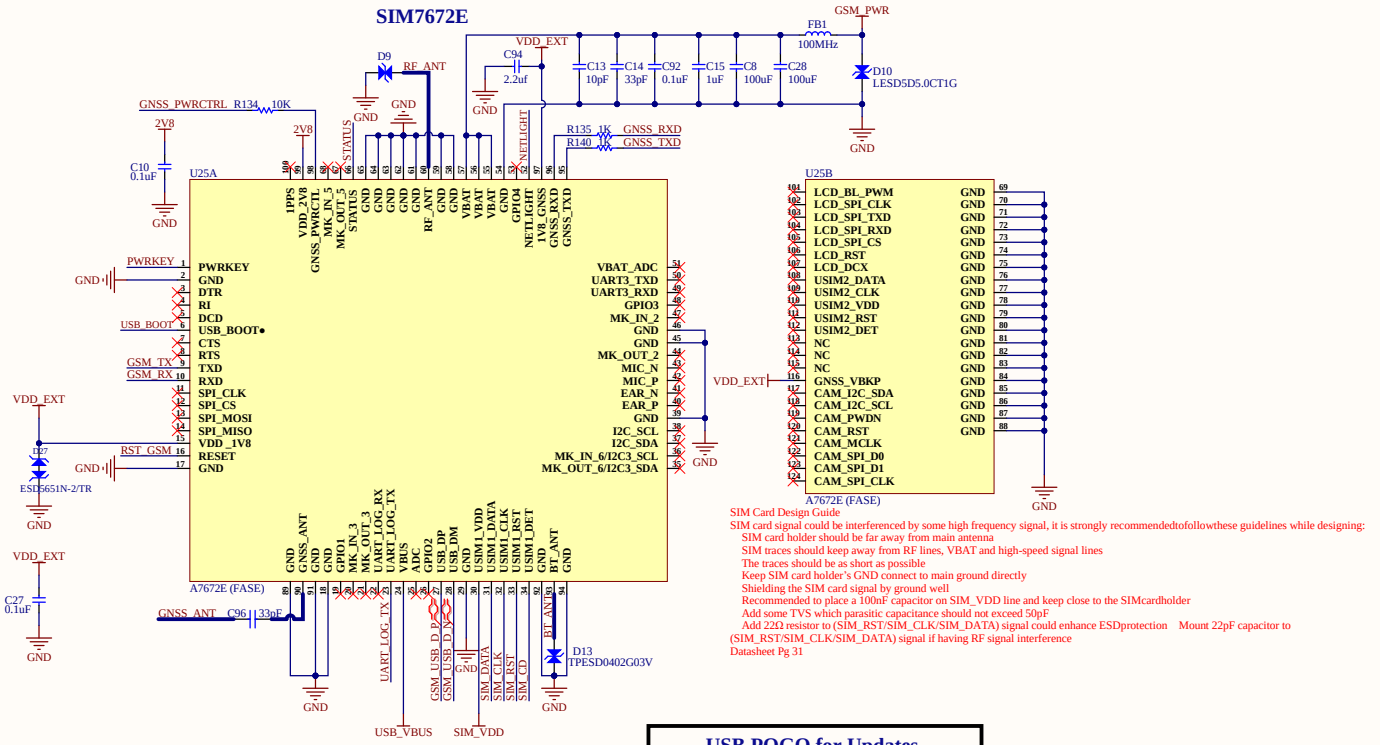




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File: GTV3.4_Board.PrjPcb	
Date: 3/7/2024	Page Num: 17
Drawn By: Huzaifa Ali	Page Total: 17

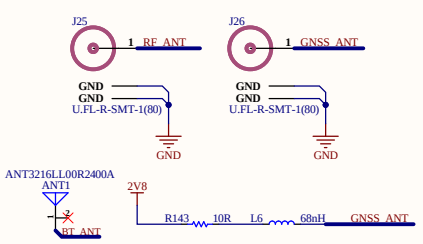
SIM7672E



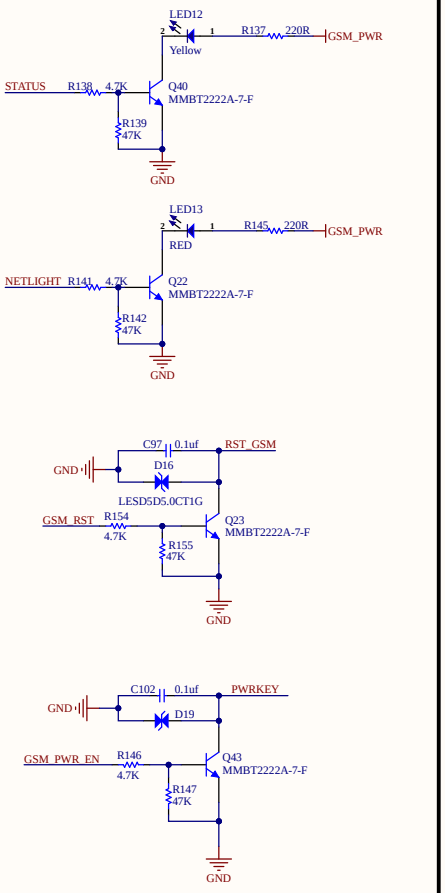
SIM Card Design Guide
SIM card signal could be interferenced by some high frequency signal, it is strongly recommendedtofollowthese guidelines while designing:
- SIM card holder should be far away from main antenna
- SIM traces should keep away from RF lines, VBAT and high-speed signal lines
- The traces should be as short as possible
- Keep SIM card holder's GND connect to main ground directly
- Shielding the SIM card signal by ground well
- Recommended to place a 100nF capacitor on SIM_VDD line and keep close to the SIMcardholder
- Add some TVS which parasitic capacitance should not exceed 50pF
- Add 22Ω resistor to (SIM_RST/SIM_CLK/SIM_DATA) signal could enhance ESDprotection Mount 22pF capacitor to (SIM_RST/SIM_CLK/SIM_DATA) signal if having RF signal interference

Datasheet Pg 31

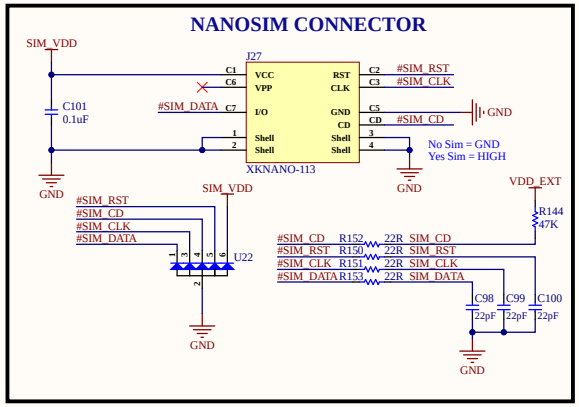
ANTENNA



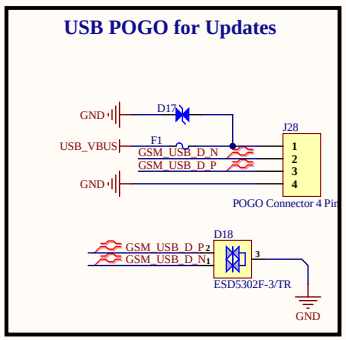
GSM_COMMANDS



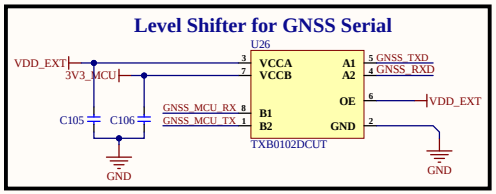
NANOSIM CONNECTOR



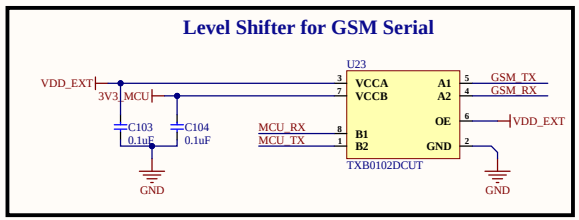
USB POGO for Updates

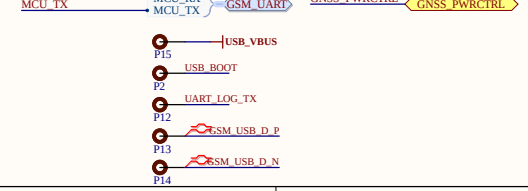


Level Shifter for GNSS Serial



Level Shifter for GSM Serial

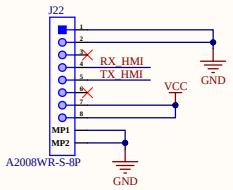




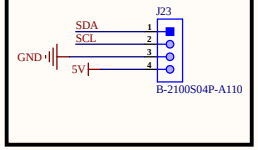
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Date: 3/7/2024	Page Num: *
Drawn By: Huzaifa Ali	Page Total: *

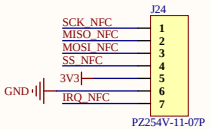
HMI INPUT STONE



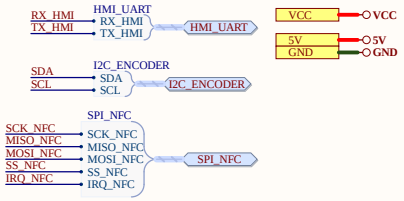
ENCODER INPUT



NFC INPUT

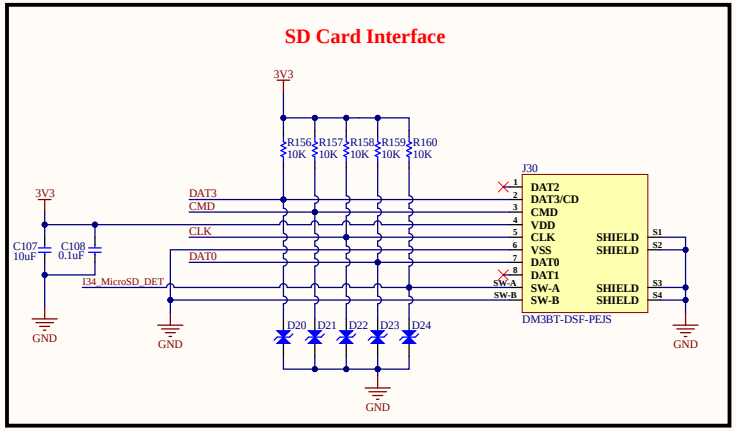


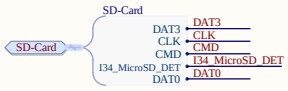
PN532 NFC Module 13.56Mhz



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Date: 3/7/2024	Page Num: *
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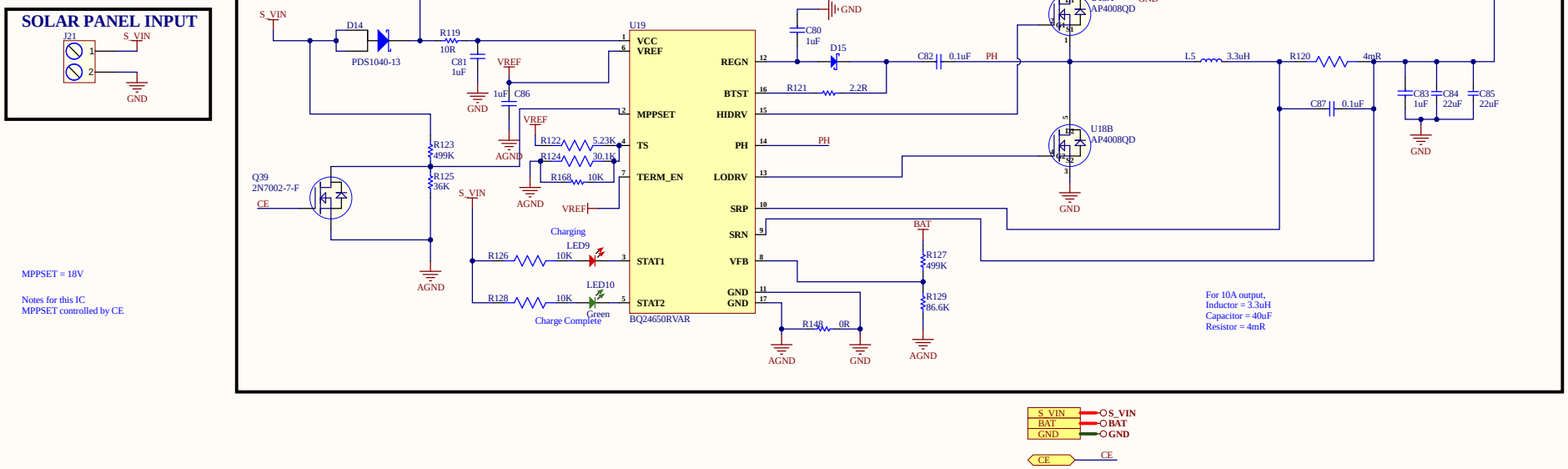






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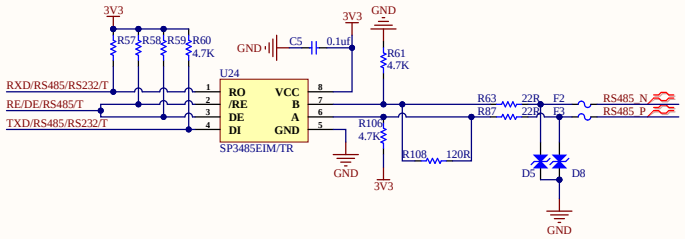
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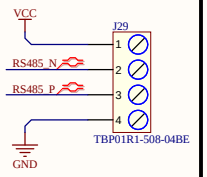
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File: GTV3.4_Board.PrjPcb	
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RS485 to UART



ULTRA-SONIC SENSOR



Part Selected:
https://www.massa.com/industrial/ultrasonic-sensors/#1501873783525-5657d00f-07c9





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